

Service creation

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This paper examines the features of an intelligent network (IN) necessary to support rapid service creation, and shows that a service creation environment (SCE) is only a part of the solution. It then outlines the service creation process and introduces three levels of service creation tool. These tools are then positioned within the BT architecture for service creation. Finally it takes the concept of IN service creation and explains how this is being expanded into total service creation, which covers all aspects of creating a new service and not just the network aspects.

1. Introduction

Most diagrams of an intelligent network (IN), for instance the one in Fig 1, show a service creation environment (SCE) as one of the major components of an IN [1]. However, while an SCE is a necessary and key component, it is not sufficient in its own right and there needs to be many other features present to really attain the objective

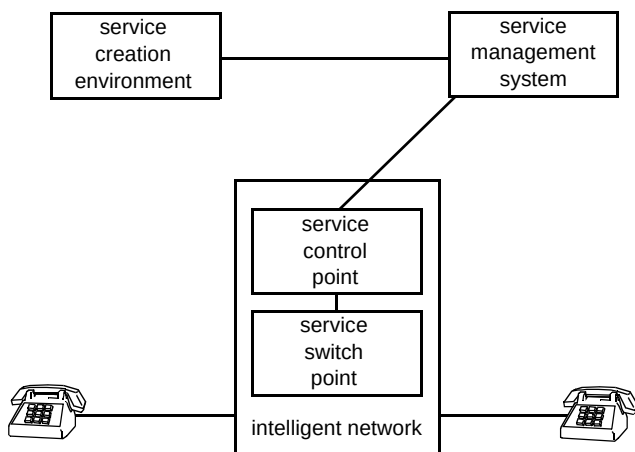


Fig 1 Simple IN architecture.

of rapid service creation. This paper will examine several such features and explain how each contributes to the goal of faster service delivery. The overall process for rapid service creation is then presented and from this it becomes clear that SCE is a hierarchy of tools rather than a single environment, with each toolset being targeted at a particular class of

user. These tools are then positioned within the BT service creation architecture. Finally, the paper will present the vision for total service creation, which takes the concepts of IN service creation but applies them to all of BT's operational support systems (OSS) [2]. These systems provide functions such as managing the underlying network, allowing orders to be placed and tracked, allowing faults to be reported and monitored, and providing the ability to charge and bill for new services.

2. IN service creation

This section examines several of the attributes that are required in an IN to accelerate the creation of new services. They are:

- functional separation — this separates the basic core functions of real time call switching from the customer and service specific aspects, so that the latter can be changed more easily,
- portable software environment — enabling services to be developed once and then run on different vendor SCPs,
- generic building blocks — building services from common modules to allow considerable reuse and hence speed of development,
- service logic programs (SLPs) — a simple language for specifying the linkages between building blocks, usually produced by service creation tools,

- graphical creation tools — these tools enable services to be rapidly created by ‘on-screen’ manipulation of icons which represent the generic building blocks,
- service and network simulators — when a service has been created, it can immediately be simulated on a range of simulators that check its functionality, performance, cost, etc,
- on-line deployment — when a service is ready for deployment it can be electronically sent to the network and the appropriate management systems, from the service creation tool.

2.1 Functional separation

This is the essential concept of the intelligent network, creating a network intelligence layer of SCPs, separate from the basic switch layer of SSPs. The latter have severe real time and performance constraints and because there are so many SSPs in the network they need to be highly optimised. Therefore the aim here is to produce a very stable environment that can be exhaustively tested and refined. SSPs need store only relatively small amounts of service or customer specific data.

In contrast, the SCP is designed specifically to run a wide range of network services. It is also designed to handle large amounts of volatile customer data.

2.2 Portable software environment

Technical developments and commercial considerations will invariably lead to changes in SCP hardware. If all existing applications had to be rewritten for each new platform, this would clearly limit the growth of new services. Therefore the approach taken is to produce the applications for a portable software environment that totally isolates them from the physical network and hardware implementation. This approach has similar advantages to using a common operating system across a range of machines.

2.3 Generic building blocks

Together with the service creation tools themselves, the use of reusable generic building blocks is the real key to rapid service creation. In the general world of computing, much work has been done on trying to achieve significant software reuse because the benefits are enormous. For example, reusing a program three times effectively triples productivity and furthermore on the second and subsequent time it is used, the lead time is virtually zero.

Despite these huge benefits, general software reuse is still rare because it is difficult to realise. However, within the fairly closed domain of the SCP, where the discrete operations of the underlying network are well understood; it is very possible to build generic pieces of software to drive these underlying operations. These building blocks are then called upon in many different sequences to provide the diverse range of IN services.

A typical building block might be ‘time-of-day routing’. This building block would check the user’s profile, which may state that after 6:00 pm calls to this number are diverted to a night-watchman. The building block would then check the current time and route the call appropriately.

2.4 Service logic programs

Services are created from the generic building blocks by specifying the sequences of these blocks and any conditional linkages between the blocks. This specification is frequently known as a script or service logic program and is usually generated by a service creation tool.

When this SLP is deployed into the network it needs to be ‘executed’. The module for doing this is often called a service logic interpreter (SLI) because many IN implementations use an interpreted language [3] for their SLPs. Within BT the term service engine is preferred to SLI, as the flexibility is needed to support interpreted, compiled [3] and possibly other forms of SLP.

2.5 Graphical creation tools

Service creation environments (SCEs) or tools represent the user interface into service creation. At present most attention is focused on graphical tools, although in the future other technologies such as expert systems may be used.

The basic concept of a graphical SCE is that icons are used to represent the generic building blocks described in section 2.3. The human service creator then drags and drops these icons on the screen and sets up linkages between the icons to represent the flow of the service. These links or flows can also be made conditional upon certain events occurring, e.g. upon a certain number being dialled. This on-screen representation of the service can then be converted into a script or SLP for deployment down to the network SCP. The on-screen display and underlying representation of these tools varies from vendor to vendor. The BT toolset uses the concept, though not the notation, of Fenton and Whitty [4] flowgraphs which are based on graph theory. The Fenton-Whitty theory is also explained and

extended in some of the Esprit II Cosmos project documentation [5, 6].

In addition to a service editor, there are usually many other functional modules present in an SCE. Some of these are support functions to do with security access and configuration control, etc. Some are simulators and modules to control the release of services, and these are covered in the next two sections. Other modules may be concerned with particular resources in the network. For instance, there are specialised modules to handle the recording of speech messages that may be played by an intelligent peripheral during a voice dialogue with a customer.

So far this section has assumed there is a single SCE, albeit with some modules being optional. In practice, the requirements of different users are such that no single tool would be adequate or appropriate. Therefore, BT has developed an architecture that identifies three distinct types or levels of service creation tools. The rationale for these levels is presented in section 3 and they are described in section 4.

2.6 Service and network simulators

Once a new service is created, it needs to be 'tested' and its impact on the underlying network and other services need to be assessed. A range of tools or simulators can be provided in an SCE to automate and support these processes.

Some of the functions available are:

- functional simulation — this provides a complete emulation of the service, with all the correct look and feel, through some customer premises equipment (CPE) attached to the simulator,
- service logic animation — this enables the service logic to be 'executed' within the simulator with the various icons and links on the screen being highlighted as they are activated,
- network animation — a graphical representation of the network can be displayed, with icons of phones, etc, in various locations, phones at various sites then being stimulated and the resultant traffic flows displayed,
- network performance simulation and analysis — using the same underlying model of the network, conventional traffic theory algorithms can be used to look for bottlenecks and estimated network delays,
- cost and revenue analysis — these tools use estimates of network resources utilisation to either cost a new service or make revenue predictions for the service,

- feature interaction analysis — feature interaction is a big problem for IN and not all of it can be prevented in an SCE; however, through various analyses it is possible to detect and reject some erroneous interactions at this stage.

2.7 On-line deployment

Once the service is created and tested it is ready to be deployed. The bulk of the software for a service is already present in the network in the form of generic building blocks. Therefore all that requires deploying is the relatively small service logic program, which can easily be sent electronically.

In theory a new service could instantly be deployed on to an entire network. In practice it is usual to initially deploy the service into a stand-alone node or small sub-network so that the service can be further tested and trialled before major roll-out.

3. Service creation process

This section describes how these individual IN features can be harnessed together to produce a rapid service creation process. Firstly, it is useful to consider the traditional life cycle for putting a new service into a network, which is:

- design and enhance the physical network — this prepares the underlying network to support the new service, items such as new types of intelligent peripheral perhaps being required for addition to the network (alternatively new switch functionality or capacity may be required),
- develop the basic software for the new service,
- establish the packaging and marketing aspects of the service (How are various combinations of the service to be sold? What is the pricing structure? What are the service level agreements and the arrangements for ordering?),
- if necessary and possible, tailor the service to a particular customer.

Each of these activities can be automated to some extent but the real key to rapid service creation is to do the early steps in advance of the service being identified or required. Each of these can be pre-provisioned as described below.

- Enhance the physical network

Providing a rich functionality IN means that many new services can be created without requiring new network upgrades. Of course, this is the main justification for an IN.

- Develop the basic software

This is where the concept of building blocks comes in. Faced with the problem of building a plastic doll's house but without yet knowing the exact requirement, one could produce a set of Lego™ bricks from which any doll's house could be built, once the specification is known. If chosen appropriately, these same bricks could also be used to build other toys such as boats, which are where the real benefits of the building block approach lie.

- Establish the packaging and marketing aspects of the service

Remaining with the Lego™ analogy. Not only can the bricks be made in advance, they can often be packaged and priced in advance, allowing the customer to specify their own toy from a standard kit. For instance, a certain package may be sold as a car-making kit and would contain some standard bricks and a few 'special' building blocks such as wheels.

- If necessary and possible, tailor the service to a particular customer

If all the above has happened, then it should be possible for a customer, or a marketing person acting on behalf of a set of customers, to create a service very quickly. Even this process could be accelerated by pre-linking some of the blocks into an example service. This would be the Lego™ equivalent of selling a package with most of the bricks already joined together to form a basic car.

The first item is really network design rather than service creation but is listed here for completeness. The other items are known as the three levels of service creation. — levels 1, 2 and 3.

So far this discussion has centred on making functionality available to the next level in the process and this is shown on the right hand side of Fig 2. However, in some cases the 'package' available at level 3 will not be adequate to build the service required. In this case, action is required at level 2 to provide the missing capability. This may mean re-packaging an existing piece of functionality or it may require a new building block to be developed at level 1. This process is shown on the left-hand side of Fig 2. In the worst case one has to cascade all the way down the left-hand side of the diagram, in which case the process defaults to the traditional development cycle described at the start of this section.

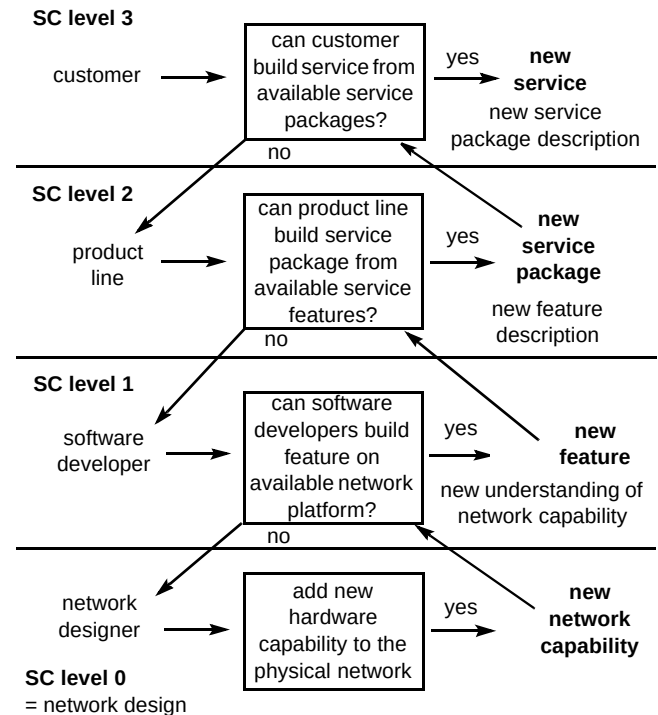


Fig 2 Service creation process.

4. BT's service creation architecture

BT has developed an architecture for service creation that reflects the service creation process just presented. This architecture is shown in Fig 3. Space here does not permit an examination of the service and network management systems. However, for the purposes of this discussion, they are responsible for the storage and distribution of the various outputs from the SCE tools and their deployment into the network.

Within this architecture there are specialised SCE tools for each service service creation level which are now examined in more detail.

4.1 SCE level 1

These tools are used by skilled software developers to produce the generic building blocks. They are generally based around conventional computer-aided software engineering (CASE) tools [7]. However, they need particularly good configuration management controls.

In addition to downloading the actual building blocks to the target systems, they must also pass a representation of the building block to the Level 2 tool. This representation covers:

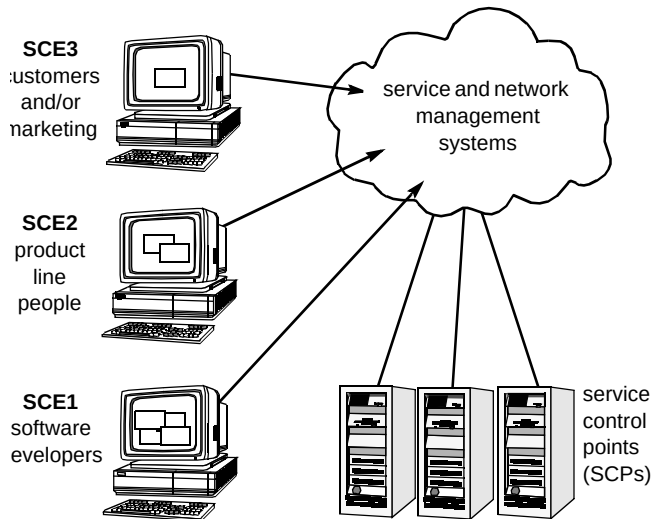


Fig 3 Service creation IN architecture.

- a textual description of the block, for information purposes and as help text,
- its on-screen presentation, i.e. the icon that schematically represents the function,
- the rules governing interactions with other building blocks,
- the data associated with the building block — these will eventually need to be provisioned for a particular customer.

4.2 SCE level 2

These tools are used by product line developers or agents acting on their behalf.

From a purely IN service perspective, an SCE2 tool must provide the following functions:

- assemble a basic service, by dragging and dropping icons on a screen and linking them as described in section 2.5,
- define how much of the service flow can be altered at SCE level 3, i.e. by a customer — the rest of the service is then effectively fixed for all customers,
- preset some of the data associated with the service with default values and again define which of these items a customer will be allowed to overwrite.

With the advent of total service creation, as described in the next section, it is now possible and appropriate that SCE2 should do more. In particular, it is the obvious place to define the packaging and marketing information,

described as service creation level 2 activities in section 3. Examples of these additional functions are:

- define how orders may be placed and tracked,
- define how the service may be charged and billed,
- define how network and customer reported faults may be handled,
- define what SLAs and performance measurements are allowed.

4.3 SCE level 3

These tools are used by customers directly, by BT operators acting on behalf of specific customers or by marketing people acting on behalf of groups of customers. To support these various scenarios, the SCE3 functionality can be provided on a range of hardware, from personal computers (PCs) to sophisticated Unix workstations.

The functions supported through an SCE3 are:

- tailor the service — within the constraints set at SCE2, the user may be allowed to change the flow or structure of their particular version of the service,
- provision some of the service data with customer specific values.

5. Total service creation (TSC)

In order to launch a new BT Service it is necessary to ensure that:

- the network can provide the service,
- there are operational support systems in place that can manage the new service.

The problems of deploying new services in the existing network are well known. All switch vendors need to agree on the new service and build it into their development plans. BT then needs to deploy it, possibly into all exchanges in the network. As a result many new services are deployed on overlay networks. However, this brings problems of network management and interworking, and loss of economies of scale.

Even when an existing network can be used for a new service, there are similar problems at the service management level — for, inevitably, the existing management systems will need modification to support the new service. However, these new requirements are not specified in any standard format and are often not rigorous. This wastes time and effort in interpreting these new requirements. Also many of the current systems are difficult

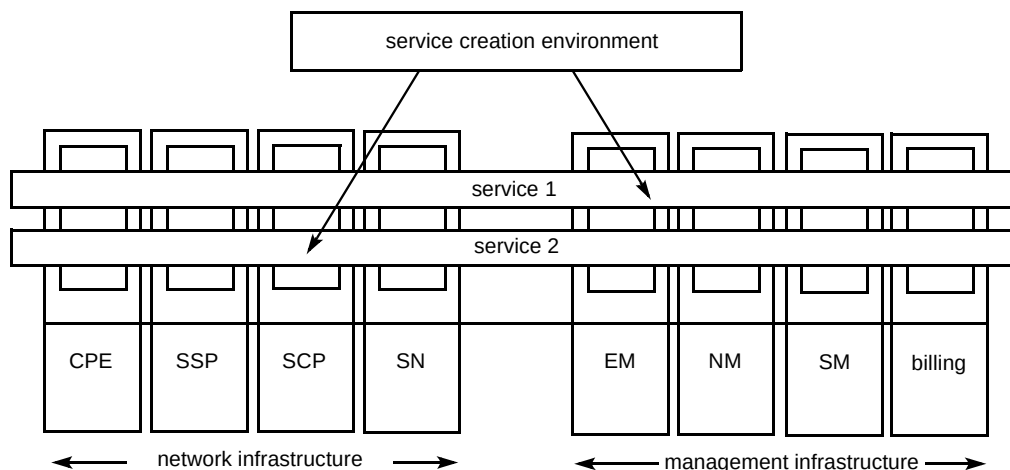


Fig 4 Total service creation concept.

to change and have a backlog of other requirements to satisfy. The net effect is that changes often cannot be made quickly enough and this leads to new management systems being introduced as interim solutions. This effectively creates an 'overlay management system', which has similar problems to an overlay network, namely lack of integration and the sheer cost and logistic problems of supporting so many diverse systems.

Intelligent networks are generally seen as the solution to the first problem, with IN service creation offering the prospect of dramatically shorter time-scales for putting a new service into the network. However, an IN does not help produce the corresponding operational support functions required to manage each new IN service.

Total service creation¹ proposes applying many of the concepts described in section 2 to the operational support system domain. Some of these concepts are now being employed in some of the new management system developments but not in any structured way. In particular, the use of graphical tools and automatic deployment is rare. Total service creation provides a framework for producing operational support functionality in a systematic way. The concept of total service creation is shown schematically in Fig 4.

6. Conclusions

Total service creation takes the concepts that allow rapid network service creation in an intelligent network and generalises them to propose 'rapid management service creation' in an 'intelligent management domain'.

When fully implemented, total service creation will enable a service designer to sit at a service creation environment workstation and be taken through a process that defines all aspects of the service and its operational support. This new service could then be simulated and tried on a reference model or a subset of the main network. Finally, given the necessary authority, this workstation could deploy this new service into the intelligent network and simultaneously upgrade all the appropriate management and billing systems to support this new service.

Adopting the approach of total service creation will enforce a common method of defining new services and the associated management features.

Total service creation will enable many services to be provided in a few weeks or months rather than years. This will have large savings in terms of the costs and number of people involved in developing the current range of operational systems. However, teams will still be required to develop and continuously enhance the systems that support total service creation.

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Graham Turner graduated from the University of London in 1973, with a first class honours degree in physics. He joined BT that same year, working in their Research Centre.

Here he co-developed the software for the world's first Viewdata system, which won the prestigious MacRobert Medal. He then worked on all aspects of videotex, including a period working with CCITT to define the emerging international standards. By 1981 he was managing the videotex development section, which won the Queen's Award to Industry for technical innovation. He then managed a number of product developments in telematics, office automation, network and service management which included the development of the Advanced Service Management System for BT's international virtual network.

He is currently the service creation portfolio manager, responsible for developing BT's service creation tools and associated network components.